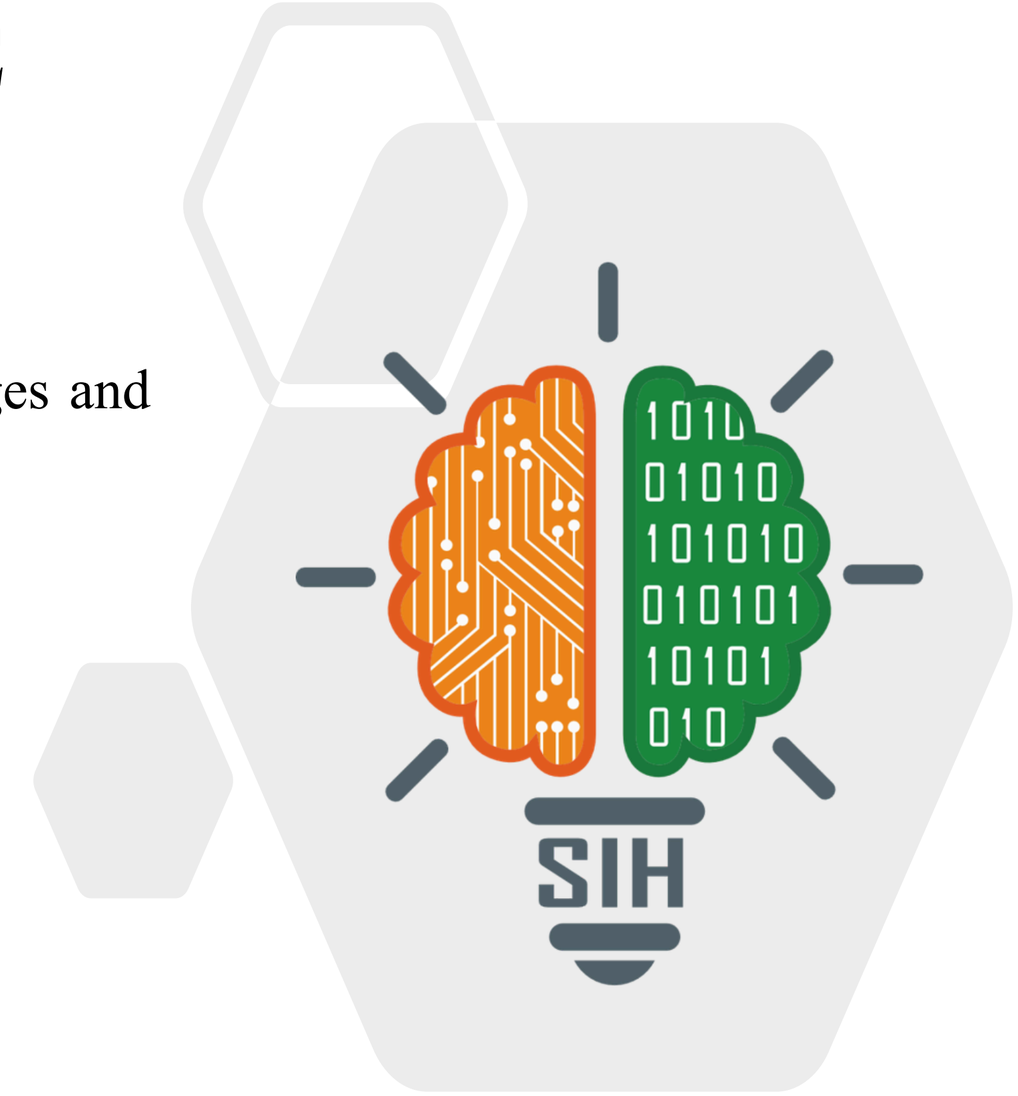
Krishi Niti

SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID** – 26132
- **Problem Statement Title-** Strengthening market linkages and price discovery for farmers
- **Theme-** Agriculture, FoodTech & Rural Development
- **PS Category-** Software
- **Team ID-** 121083
- **Team Name :** NeuroBytes



KRISHI NITI(कृषि NITI)

शेतापासून बाजारापर्यंत, सुरक्षित आणि स्मार्ट प्रवास

NeuroBytes

Krishi Niti

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Problems faced by farmers in market

- 6 Critical Gaps Across Pricing, Logistics & Trust → See How We Solve Them

Proposed solution: Krishiniti Web/App

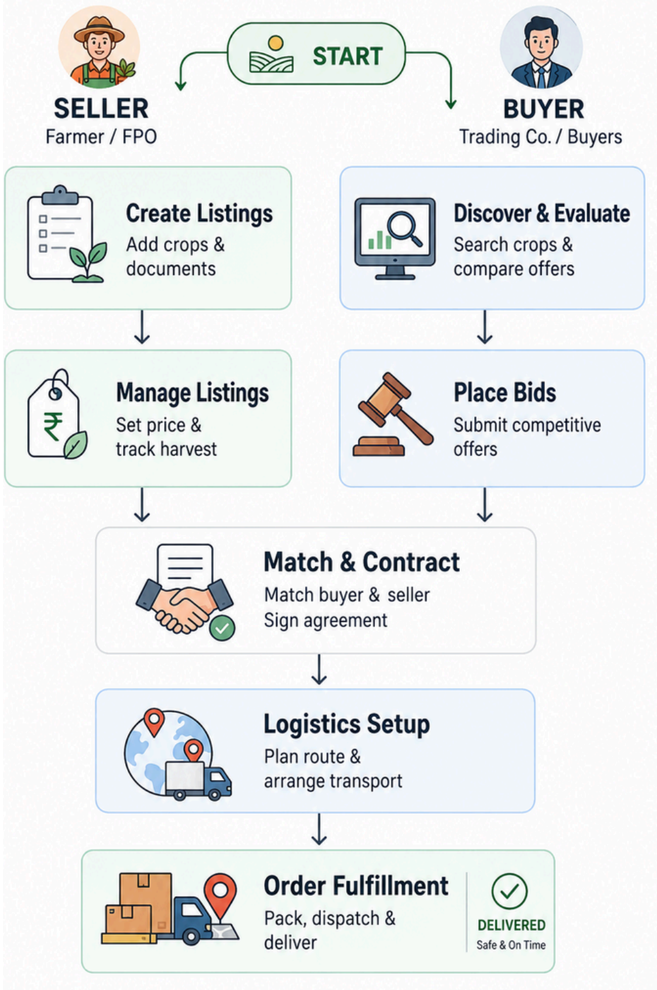
A Decentralised, Multilingual platform which uses AI and blockchain to connect farmers directly with buyers

Existing solutions

- eNAM/Agmarknet — price discovery only, no bidding/logistics/on-chain trust
- DeHaat — advisory & inputs, no bidding or escrow
- Ninjacart — fixed-price B2B, no peer bidding or blockchain trust

Unique Features Addressed:

- Bidding System
- Immediate Warehouse Support
- AgriScore to measure reliability
- Centralised Main Price feed page
- Decentralised network (private) reducing transaction cost
- Proper transaction audit on our blockchain network



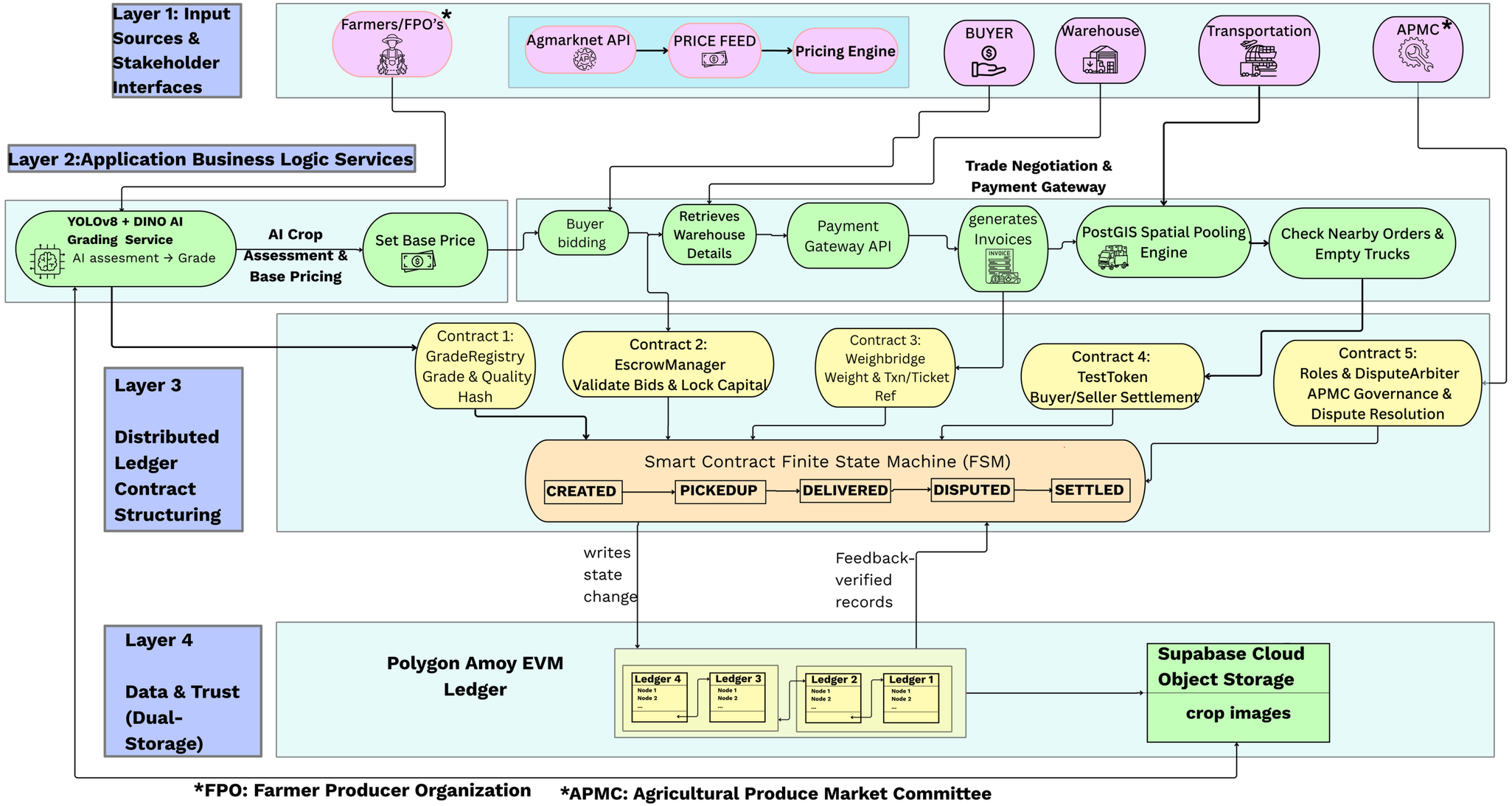
Links [Proposal](#) [Video](#) [Website](#)

TECHNICAL APPROACH

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*FPO: Farmer Producer Organization

*APMC: Agricultural Produce Market Committee

Tech stack

01 Frontend
- Next.js & React
- Tailwind CSS & TypeScript

02 Backend
- Python (FastAPI)
- Pydantic & Uvicorn

03 AI & Vision
- YOLOv8 (Quality Assay)
- OpenCV & NumPy

04 Database & Geo
- PostgreSQL (PostGIS)
- Redis

05 Blockchain
- Polygon Amoy
- EVM-compatible chain

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FEASIBILITY AND VIABILITY

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Analysis Of Feasibility

Technical:



- Role-Based Dashboards
- Government & System Integration
- Phased Deployment & Training
- Automated Operations & Workflows

Economic:



- Cost-Efficient ROI: 8–10 mo. break-even
- Centralized Process Efficiency: ~30% faster
- Scalable Cloud Infrastructure: ₹12–15K/mo (10K users)
- Low Cost of Ownership: ₹40–60/farmer; <2s latency
-

Operational:



- Scalable Cloud Architecture
- Automated Real-Time Analytics
- Secure Enterprise Control
- Real-Time Logistics Tracking

Gaps Identified VS Our Solution

Krishi Niti: Data-Driven Architectural Alignment

Gaps Identified		Our Solution
Manual & Inconsistent KYC	01	KYC with Auto-Validation
Limited Credit Risk Assessment	02	Real-Time Inflation-Adjusted Price Feed
Inefficient Agri-Market Pricing	03	Auction Pool with 30-Sec Comparative Updates
Opaque Bidding Processes	04	Geo-Spatial Warehouse Capacity Optimizer (A vs B)
Sub-Optimal Transport Logistics & Routing	05	Multi-Leg Route Planning & Transport ID Tracking
Delayed Escrow Fund Release	06	Multi-OTP Based Escrow Workflow
Poor Supply Chain Quality Control	07	Mandatory Validation
Manual Offboarding & Strike System	08	Integrated Grievance & Strike Redressal

eNAM/Agmarknet solve price discovery; none combine bidding, escrow, and on-chain trust in one flow — that's the gap Krishi Niti fills.

IMPACT AND BENEFITS

KRISHINITI: Problem, Feature & Benefit Alignment

⚠ MARKET PROBLEMS ⚠

Unfair Pricing:
Predatory Middleman Manipulation

Information Asymmetry:
-> Lack of Real-Time Mandi Rates

Post-Harvest Losses:
-> Inadequate Storage & Transport

Unreliable Buyers:
-> Payment Delays & Default Risks

Opaque Records:
-> Payment delays + manipulated paper audit trails.

Digital Illiteracy / Tech Barrier
-> Complex Web Interfaces

PLATFORM UNIQUE FEATURES

AI-Powered Bidding System:
-> Dynamic Fair Price Discovery

Integrated Price Feed:
-> Transparent Regional APMC Data

Immediate Warehouse Support:
-> Verified Storage & Cold Chain

AgriScore (AI Trust Index)
-> Verified Buyer & Seller Rating

ChainLog (AI Audit Ledger)
-> Immutable On-Chain Records

Blockchain Audit Ledger:
-> Immutabie On-Chain Records

WhatsApp Vernacular Chatbot:
-> Direct Text Chat Produce Listing

BENEFITS & SOLUTIONS

Fair Market Realization:
Maximized Farmer Revenue

Informed Selling:
Zero Exploitation from Price Gaps

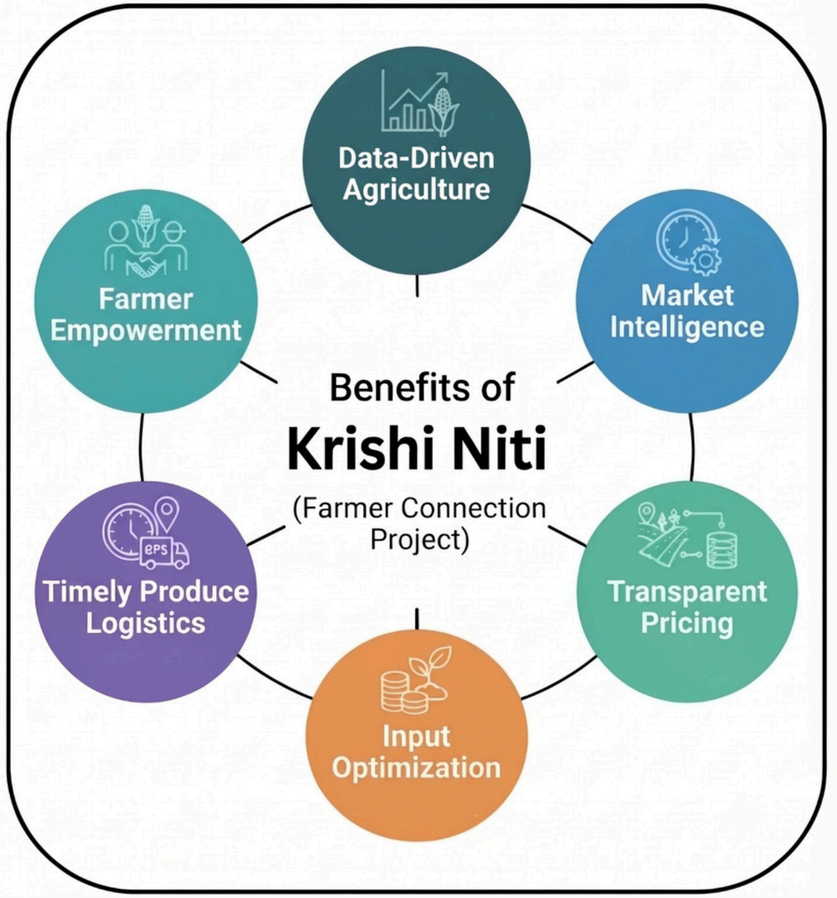
Spoilage Reduction:
Preserved Crop Shelf-Life

Trusted Counterparties:
Vetted Trading Partners

100% Traceability & Integrity:
Tamper-Proof Digital Slips

Multi-OTP Based
Secure Escrow Workflow

Mandatory Validation
Integrated Grievance & Strike Redressal



आज बेचें या कल? KrishiNiti बताएगा सही पल.

RESEARCH AND REFERENCES

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- built 30-sec comparative feed

[3] Government of India, Open Government Data (OGD) Platform, Daily Market Prices of Commodities (Mandi) Dataset. Available: https://data.gov.in

- informed AgriScore trust index

[4] National Informatics Centre, AGMARKNET Web Based Application — User Manual. Available: https://agmarknet.gov.in (User Manual section)

- automated price ingestion via API

[5] Maharashtra State Agricultural Marketing Board (MSAMB), eNAM Integration & APMC Details, Maharashtra. Available: https://www.msamb.com

- shaped decentralized APMC nodes

[6] Ministry of Agriculture & Farmers Welfare, eNAM Interstate Trade Statistics and Progress Reports. Available: https://enam.gov.in/web/about-e-nam

- standardized grading (YOLOv8+DINO)

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[2] IBMRD's Journal of Management & Research, Digital Transformation in Agricultural Marketing: Advancing India's Agri-Trade Ecosystem, Vol. 14, Issue 1 (March 2025). Available: https://ibmrdjournal.in/index.php/ibmrd/article/view/174290 - added PostGIS route optimization

[3] IRJMETS, AgriLink: Direct Market Access for Farmers — A Mobile Application Approach. Available: https://www.irjmets.com/upload_newfiles/irjmets70500180613/paper_file/irjmets70500180613.pdf - added WhatsApp vernacular chatbot